

21 Batch reactor energy balance

Why are we learning this?

In previous lectures, we developed energy balances for CSTRs and tubular reactors. Today we will derive the energy balance for a batch reactor. Because a batch reactor contains no inlet or outlet streams, the energy balance takes on a particularly simple and physically intuitive form.

By the end of today's lesson, you should understand:

- How the batch reactor energy balance is derived.
- Why batch reactors have no energy transport by mass flow.
- How heat transfer and heat of reaction affect reactor temperature.
- How to write the batch reactor energy balance in terms of temperature.
- How to write the batch reactor energy balance in terms of conversion.
- The special case of an adiabatic batch reactor.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
► Energy Balances ◀
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea: Batch Reactors Have No Inlet Streams and No Outlet Streams So the energy balance focuses on what happens inside the reactor.

Let's derive the energy balance for the batch reactor. Starting from the general energy balance equation:

No inflows or outflows in a batch reactor, so

What Makes the Batch Reactor Special?

Unlike CSTRs and PFRs, a batch reactor has no material entering or leaving during operation. As a result: No Energy In By Mass Flow, No Energy Out By Mass Flow

The reactor temperature changes only because of:

- Reaction
- Heat transfer
- Shaft work

This makes the energy balance conceptually simpler.

$\hat{E}_{sys}$ is

$\hat{E}_{sys} =$

Simplifying, $\hat{E}_{sys} =$

What Is the System Energy?

The system energy represents the total energy contained within the reactor contents. As reaction proceeds:

- Species amounts change.
- Temperature changes.
- The total energy of the reactor contents changes.

The energy balance tracks those changes over time.

$$U_i = H_i - PV_i$$

Since we ultimately seek $\Delta\hat{E}_{sys}$, we can neglect PV since

Why Does PV Disappear?

The complete thermodynamic expression includes both internal-energy and pressure-volume contributions. For the purposes of this reactor energy-balance derivation, the pressure-volume contributions do not significantly affect the temperature calculation being developed. Removing those terms simplifies the derivation while preserving the essential physics.

Now, $\hat{E}_{sys} =$

Hence $\frac{d\hat{E}_{sys}}{dt} =$

Applying the

$$\frac{d\hat{E}_{sys}}{dt} = \left[\sum N_i \frac{dH_i}{dt} + \sum H_i \frac{dN_i}{dt} \right]_{sys}$$

Why Are There Two Terms?

The product rule appears because reactor energy changes for two independent reasons:

Temperature Changes

The energy of each molecule changes.

Composition Changes

The number of molecules present changes.

Energy Change = Temperature Effect + Composition Effect

Both contributions must be included in the energy balance.

Recalling that $H_i = H_i^o(T_{ref}) + \int_{T_{ref}}^T C_{P_i} dT$

Why Use Heats of Formation Again?

Heats of formation account for the chemical energy stored in molecular bonds. Without them, Bond Breaking + Bond Formation would not be represented in the energy balance. Including heats of formation ensures that reaction energy is properly captured.

$$\frac{dH_i}{dt} =$$

Where $H_i^o(T_{ref})$ is

$$\text{So } \frac{dH_i}{dt} =$$

$$\text{Then: } \dot{Q} + \dot{W}_s =$$

Recall $\frac{dN_i}{dt}$ is

and equals

Then $\dot{Q} + \dot{W}_s =$

Recalling that $r_i = \frac{v_i}{v_A} r_A$ (assuming A is the limiting reactant), $\sum H_i r_i V =$

$\sum \frac{v_i}{|v_A|} H_i =$

the energy balance becomes

Stoichiometry Simplifies the Energy Balance

Just as in mole balances, stoichiometry relates the rates of all species to one independent reaction rate. This allows the energy balance to be rewritten in terms of:

One Reaction Rate and One Heat of Reaction

rather than separately tracking every species.

$$\dot{Q} + \dot{W}_s =$$

Solving for $\frac{dT}{dt}$:

$$\frac{dT}{dt} =$$

Where $(\Delta H^{rxn})(r_A V)$ is

Interpreting the Temperature Equation

The final energy balance describes how quickly reactor temperature changes with time. Temperature changes because of:

Heat Generated By Reaction + Heat Transfer + Work

and is moderated by the heat capacity of the reactor contents. A larger heat capacity generally means temperature changes more slowly.

And $\dot{Q}$ is

$$\dot{Q} =$$

Batch reactor energy balance written in terms of X:

Why Convert to Conversion?

Most reactor-design calculations are already written in terms of conversion. Expressing the energy balance in terms of conversion makes it easier to combine:

Mole Balance + Energy Balance

within a single solution strategy.

Recall $N_i =$

Then, $\frac{dT}{dT} =$

Recall $\sum \frac{v_i}{|v_A|} C_{P_i} =$

Then, $\frac{dT}{dT} =$

Let's do some examples:

- The reaction $A + B \rightarrow C$ is carried out in a batch reactor. $C_{PA} = 35 \text{ cal/mol/K}$, $C_{PB} = 18 \text{ cal/mol/K}$, $C_{PC} = 46 \text{ cal/mol/K}$. Determine ΔC_P .

- For the same reaction, the inlet molar flowrates are 1 mol A/min, 2 mol B/min, and 0 mol C/min. Determine Θ_i .
- The reaction is carried out adiabatically. What is the energy balance?

Adiabatic Batch Reactors

In an adiabatic reactor:

$$Q = 0$$

No heat enters or leaves the reactor. As a result:

Temperature changes only because reaction occurs. This simplifies the energy balance and creates a direct relationship between conversion and temperature.

Summary and Key Takeaways

Today we derived the energy balance for a batch reactor.

Key Ideas

- ✓ Batch reactors have no inlet or outlet streams, eliminating energy transport by mass flow.
- ✓ The reactor energy changes because of reaction, heat transfer, and shaft work.
- ✓ Heats of formation must be included to properly account for chemical energy stored in molecular bonds.
- ✓ Stoichiometry allows the energy balance to be expressed using a single reaction rate and heat of reaction.
- ✓ Adiabatic reactors are special because temperature changes arise only from reaction.

Most Important Idea

Batch Reactors Have

No Energy In By Flow

No Energy Out By Flow

So Temperature Changes Are Controlled By

Reaction

+

Heat Transfer

+

Work

The batch reactor energy balance provides the temperature equation needed to solve non-isothermal batch-reactor design problems.